

EMo Learners | Premium Solved Model Exam Paper

COMPUTER ORGANIZATION & ARCHITECTURE (COA) SOLVED BANK

SISTec Academic Mid-Semester Pattern — Semester IV

Unit 1 (Basic Structure of Computer) & Unit 2 (Computer Arithmetic & Control Memory)

Subject: Computer Organization & Architecture (CS-404)

Total Solved Marks: 28 Marks

Layout Structure: Part A (MCQ), Part B (Short), Part C (Long)

Branding Highlight: EMO Learners Exclusive
Solved Paper

PART A: OBJECTIVE MULTIPLE CHOICE QUESTIONS (6 MARKS)

Instructions: Answer all twelve questions. Each question carries exactly 0.5 marks.

Q.1 What is meant by the Von-Neumann bottleneck?

- A. CPU processing cycles are too slow
- B. Shared bus restricts simultaneous data and instruction fetch
- C. Registers are too small to store data
- D. Multiplexer selection delays the ALU

Answer: B. Shared bus restricts simultaneous data and instruction fetch

Explanation: Since instructions and data share the same physical bus in the Von-Neumann architecture, the CPU cannot read an instruction and read/write data at the same instant, throttling throughput.

Q.2 In register transfer language (RTL), what does the expression 'P : R2 <- R1' represent?

- A. R1 loads R2 at all clock pulses
- B. R1 and R2 swap values on condition P
- C. If Boolean condition P = 1, R2 loads contents of R1
- D. R2 is cleared if P = 0

Answer: C. If Boolean condition P = 1, R2 loads contents of R1

Explanation: P is a control function (Boolean condition). When P evaluates to 1, the target microoperation (transferring R1 to R2) is executed at the next clock edge.

Q.3 Which addressing mode does not require a memory reference to fetch the operand?

- A. Direct Addressing
- B. Indirect Addressing
- C. Immediate Addressing
- D. Relative Addressing

Answer: C. Immediate Addressing

Explanation: In immediate addressing, the operand is explicitly defined inside the instruction word itself, requiring zero memory access cycles to fetch data.

Q.4 Which CPU register holds the address of the next instruction word to be fetched from memory?

- A. Instruction Register (IR)
- B. Program Counter (PC)
- C. Accumulator (AC)
- D. Memory Address Register (MAR)

Answer: B. Program Counter (PC)

Explanation: The Program Counter (PC) is a dedicated register that points to and tracks the next instruction's address in system memory.

Q.5 How many 8-to-1 multiplexers are required to construct a common bus system for eight 16-bit registers?

- A. 8 multiplexers
- B. 16 multiplexers
- C. 24 multiplexers
- D. 128 multiplexers

Answer: B. 16 multiplexers

Explanation: A common bus system requires exactly one multiplexer for each bit line of the register size. For 16-bit registers, we need 16 multiplexers (each being 8-to-1 since there are 8 registers).

Q.6 What is the range of signed integers that can be stored in an 8-bit register using 2's complement representation?

- A. -127 to +127
- B. -128 to +127
- C. -128 to +128
- D. 0 to 255

Answer: B. -128 to +127

Explanation: Using n bits, 2's complement represents values from $-2^{(n-1)}$ to $+(2^{(n-1)} - 1)$. For 8 bits, this yields -128 to +127, featuring a unique single representation of zero.

Q.7 In Booth's multiplication algorithm, what action is taken when the multiplier bits Q_0 and Q_{-1} are equal to '10'?

- A. Shift only (ARS)
- B. Add multiplicand to accumulator, then shift
- C. Subtract multiplicand from accumulator, then shift
- D. Clear the accumulator

Answer: C. Subtract multiplicand from accumulator, then shift

Explanation: Booth's algorithm rules specify: 00 or 11 = shift only; 01 = add and shift; 10 = subtract ($A = A - M$) and shift.

Q.8 What is the biased exponent offset value used in the standard IEEE 754 Single-Precision floating-point format?

- A. 127
- B. 128
- C. 1023
- D. 255

Answer: A. 127

Explanation: IEEE 754 single-precision format uses 8 bits for the exponent with a bias of 127, allowing representation of exponents from -126 (stored as 1) to +127 (stored as 254).

Q.9 What timing component in a hardwired control unit is used to generate sequential timing state signals T0, T1, etc.?

- A. Instruction Decoder
- B. Step Sequence Counter
- C. Control Address Register
- D. Control Memory ROM

Answer: B. Step Sequence Counter

Explanation: A step counter increments continuously on clock edges, feeding into a decoder that generates sequential timing states (T0, T1, T2) to guide microoperations.

Q.10 Where are the control words (microinstructions) stored in a microprogrammed control unit?

- A. Primary RAM
- B. CPU General Registers
- C. Control Memory (Internal ROM)
- D. Instruction Register

Answer: C. Control Memory (Internal ROM)

Explanation: In a microprogrammed control unit, control signals are pre-written as microcode and stored in a high-speed internal Control Memory ROM.

Q.11 Which microinstruction organization offers higher parallel execution capabilities and faster operating speed?

- A. Vertical Microinstruction
- B. Horizontal Microinstruction
- C. Encoded Microinstruction
- D. Indirect Microinstruction

Answer: B. Horizontal Microinstruction

Explanation: Horizontal microinstructions dedicate a separate bit to each physical control line. This eliminates decoding delays, allowing multiple microoperations to execute in parallel instantly.

Q.12 In Restoring Binary Division, what action is taken when the partial remainder in register A becomes negative (MSB = 1) after subtraction?

- A. Set quotient bit Q0 = 1
- B. Add divisor back to A to restore its state, and set Q0 = 0
- C. Arithmetic right shift A-Q
- D. Terminate the division loop

Answer: B. Add divisor back to A to restore its state, and set Q0 = 0

Explanation: Restoring division subtracts the divisor. If the result is negative, it must 'restore' the original value by adding the divisor back, setting the quotient bit Q0 to 0.

PART B: SHORT ANSWER QUESTIONS (8 MARKS)**Section 1: Unit 1 (Basic Structure of Computer)**

Instructions: Answer any two of the following three questions. Each question carries 2 marks.

Q.1 Define the Von-Neumann Bottleneck and its main causes. List two mitigation techniques.

[□ Marks: 2.0]

Answer:

The Von-Neumann Bottleneck is a throughput limitation where CPU processing speeds are severely throttled because the data/instruction throughput of the shared bus is much slower than CPU speeds.

- Causes: Instructions and data share the exact same physical memory bus. The CPU cannot perform an instruction fetch and a data read/write at the same instant.

• Mitigation Techniques:

1. **Cache Memories:** Integrating high-speed L1/L2/L3 caches inside the CPU.
2. **Harvard Architecture:** Splitting instruction caches and data caches into physically separate structures.

Q.2 Define a three-state buffer and explain its role in a common bus system.

[□ Marks: 2.0]

Answer:

A three-state (tri-state) buffer is a digital logic gate with three output states: Logic High (1), Logic Low (0), and a High-Impedance State (Hi-Z).

- Hi-Z State: In this state, the output is electrically disconnected from the wire, behaving like an open switch.
- Role in Bus System: By connecting the outputs of multiple registers to the same bus wires via tri-state buffers, we can ensure that only one register's enable line is active to 'drive' the bus at any single moment. This prevents short circuits and bus contention.

Q.3 Compare Direct and Indirect addressing modes with suitable examples.

[□ Marks: 2.0]

Answer:

- Direct Addressing Mode: The address field of the instruction contains the actual effective address of the operand in memory.

Example: `ADD 250H` fetches the operand directly from RAM address `250H` and adds it to the accumulator.

- Indirect Addressing Mode: The address field of the instruction contains a pointer address that stores the actual effective address of the operand.

Example: `ADD [250H]` looks at address `250H` to read a pointer `500H`, then fetches the actual operand from address `500H`.

Section 2: Unit 2 (Computer Arithmetic & Control Memory)

Instructions: Answer any two of the following three questions. Each question carries 2 marks.

Q.4 Differentiate between Hardwired and Microprogrammed control units.

[□ Marks: 2.0]

Answer:

- Hardwired Control Unit: Timing and control signals are generated directly by physical combinational logic gates, decoders, and step counters. It is extremely fast but rigid and difficult to modify.
- Microprogrammed Control Unit: Timing and control signals are stored as binary words (control words) inside an internal Read-Only Control Memory ROM. It is slower due to memory access latency but **highly flexible** and easy to modify.

Q.5 Compare horizontal and vertical microinstruction organizations, detailing their advantages and disadvantages.

[□ Marks: 2.0]

Answer:

- Horizontal Microinstruction: Every control bit in the microinstruction represents a physical control signal directly without decoding.
 - Advantages: Highly parallel (can trigger many microoperations at once) and fast (skips decoder delays).
 - Disadvantages: Wide control word width, requiring massive, expensive Control Memory.
- Vertical Microinstruction: Control signals are grouped and encoded into narrow bit fields, which are decoded by external decoders.
 - Advantages: Narrow bit width, saving substantial control memory space.
 - Disadvantages: Slower due to decoder circuit propagation delays.

Q.6 Explain the concept and advantages of a biased exponent in IEEE 754 floating-point representation.

[□ Marks: 2.0]

Answer:

The standard IEEE 754 format uses a biased exponent (bias of 127 for 8-bit exponents) to allow representation of both positive and negative powers of 2 without requiring a sign bit for the exponent field.

- Simplification of Comparisons: Representing exponents as unsigned positive numbers (ranging from 1 to 254, corresponding to actual exponents -126 to +127) allows high-speed hardware comparators to compare

floating-point magnitudes directly using standard integer comparison logic.

PART C: LONG ANSWER & TRACING QUESTIONS (14 MARKS)

Set 1: Unit 1 (Basic Structure of Computer)

Instructions: Answer any one of the following two questions (with internal choice). The question carries 7 marks.

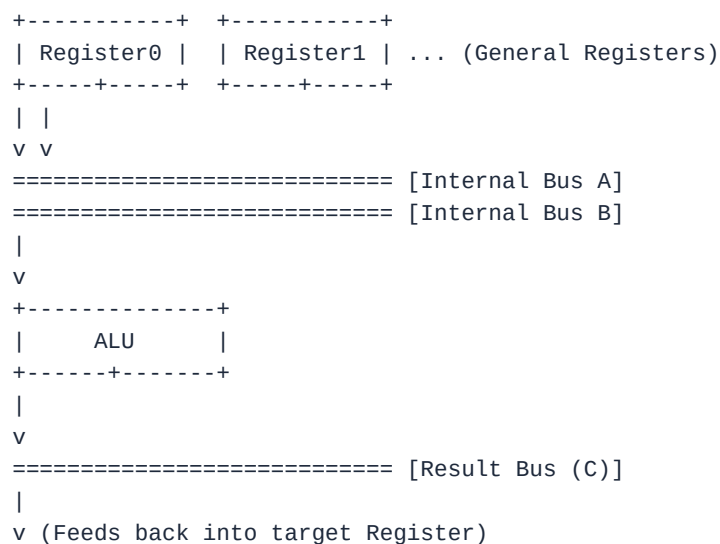
Q.7 [a] Explain the general register organization of the CPU with a neat block diagram and detail the step-by-step role of ALU and internal buses in executing $R1 \leftarrow R2 + R3$.

[□ Marks: 7.0]

Answer:

In a General Register Organization, multiple registers are connected through a common internal bus system feeding into a central ALU.

1. Bus-based CPU Register Organization Schematic:



2. Step-by-Step Execution of $R1 \leftarrow R2 + R3$:

To execute the addition microoperation in a single clock cycle, the control unit coordinates the internal buses as follows:

- **Select Source Operands (T1):** The control unit configures the multiplexer select lines ****SELA**** to place the contents of R2 onto Bus A, and ****SELB**** to place the contents of R3 onto Bus B.
- **ALU Operation Selection (T2):** The control unit configures the ALU operation selector ****OPR**** to perform addition.
- **Write Back Result (T3):** The addition sum is placed on the Result Bus C. The control unit configures the destination decoder ****SELD**** to enable the ****Load (LD)**** line of register R1. On the next active clock edge, R1 locks in the sum.

----- OR -----

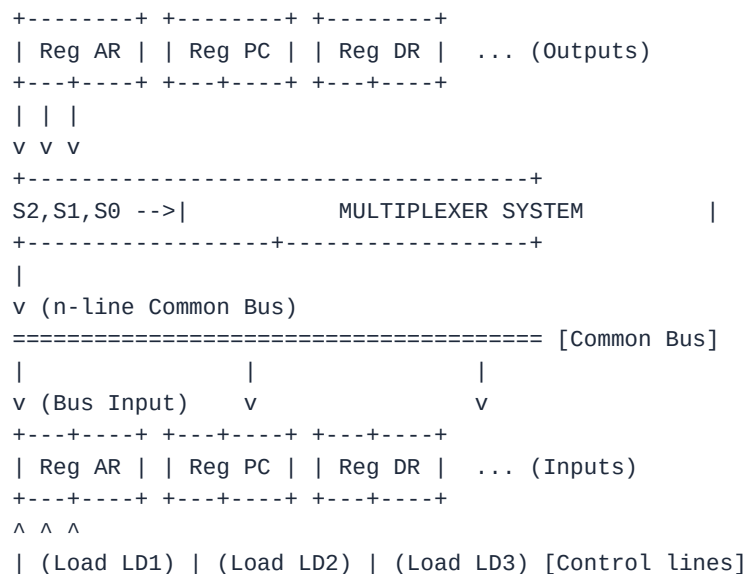
Q.7 [b] Explain the connection of registers in a basic computer to a common bus system using multiplexers. Detail the sizing rules (number of MUXes, MUX size) for k registers of n bits each with a diagram.

[□ Marks: 7.0]

Answer:

Rather than wiring every register output to every other register input, outputs are multiplexed onto a shared Common Bus System.

Common Bus Architecture Multiplexing Schematic:



Multiplexer Sizing and Construction Rules:

To construct an n -line common bus for k registers of n bits each:

- Number of Multiplexers: Exactly n multiplexers (one MUX for each bit line of the bus). MUX 0 handles bit 0 of all registers, MUX 1 handles bit 1, and so on.
- Size of each Multiplexer: Each must be a k -to-1 Multiplexer since we have k candidate registers to select from.
- Selection Lines (s): The selection inputs of all n MUXes are tied in parallel. The number of selection lines is $s = \lceil \log_2 k \rceil$.
- Selective Load (LD): The bus lines are wired directly to the inputs of all registers. To write, the control logic activates only the target register's individual Load (LD) line. On the clock edge, it loads the data.

Set 2: Unit 2 (Computer Arithmetic & Control Memory)

Instructions: Answer any one of the following two questions (with internal choice). The question carries 7 marks.

Q.8 [a] Explain Booth's multiplication algorithm. Trace step-by-step signed multiplication process to multiply (+13) and (-9) in 6-bit binary, showing Accumulator, Multiplier, and Q₋₁ registers.

[□ Marks: 7.0]

Answer:

Let us multiply $(+13) \times (-9)$ using Booth's Algorithm in 6-bit signed 2's complement representation:

- Multiplicand (M) = $+13 = 001101_2$
 - Negative Multiplicand (-M) = $-13 = 110011_2$ (2's complement of 001101)
 - Multiplier (Q) = $-9 = 110111_2$ (2's complement of 001001)
- **Registers:** Accumulator (A) = 000000 (6 bits), Q₋₁ = 0 (1 bit), SC = 6.

Cycle-by-Cycle Tracing Table:

Cycle / Action Step	Accumulator (A)	Multiplier (Q)	Q ₋₁	SC
Initial State	000000	110111	0	6
Cycle 1: Q ₀ Q ₋₁ = 10 → A = A - M ARS only	110011 111001	110111 111011	0 1	6 5
Cycle 2: Q ₀ Q ₋₁ = 11 → ARS only	111100	111101	1	4
Cycle 3: Q ₀ Q ₋₁ = 11 → ARS only	111110	011110	1	3
Cycle 4: Q ₀ Q ₋₁ = 01 → A = A + M ARS only	001011 000101	011110 101111	1 0	3 2
Cycle 5: Q ₀ Q ₋₁ = 10 → A = A - M ARS only	111000 111100	101111 010111	0 1	2 1
Cycle 6: Q ₀ Q ₋₁ = 11 → ARS only	111110	001011	1	0

Result Verification:

The final combined register contents are A Q = 111110001011₂ (12 bits signed).

Since the MSB is 1, the result is negative. Let's find its magnitude via 2's complement:

1. Invert all bits → 000001110100
2. Add 1 → 000001110101₂
3. Convert to decimal → $2^6 + 2^5 + 2^4 + 2^2 + 2^0 = 64 + 32 + 16 + 4 + 1 = 117$.

Therefore, the value is -117. Multiplying $(+13) \times (-9)$ yields -117. The trace is correct!

----- OR -----

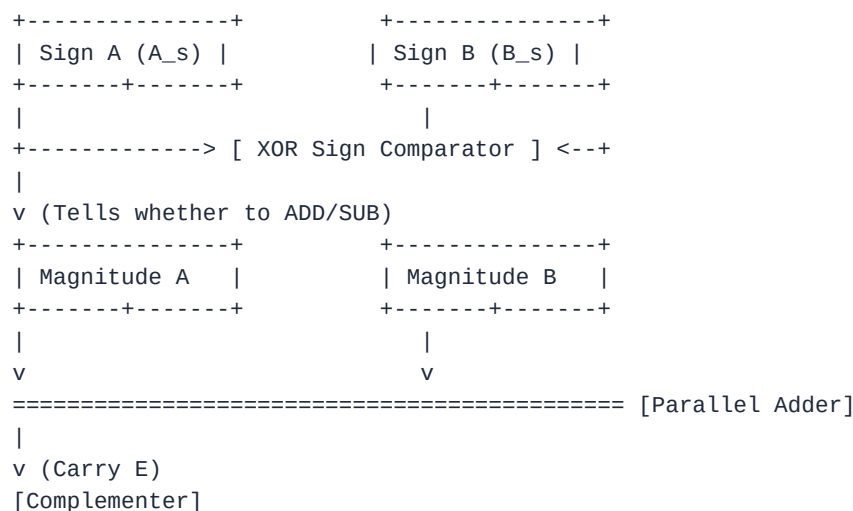
Q.8 [b] Describe the procedure for addition and subtraction for fixed point signed-magnitude numbers. Show the hardware configuration and explain overflow detection.

[□ Marks: 7.0]

Answer:

Addition and subtraction of numbers in signed-magnitude representation require comparing the signs of the operands. If the signs are identical, we add the magnitudes. If the signs are opposite, we compare magnitudes and subtract the smaller from the larger, applying the sign of the larger value.

1. Hardware Configuration Block Diagram:



2. Overflow Detection (AVF):

An overflow occurs only during the addition of two numbers of the ****same sign****. If a carry is generated out of the parallel adder (Carry E = 1), it indicates that the sum magnitude exceeds the register capacity. An overflow flip-flop (AVF) is set to 1.

- ****If E = 1:**** The overflow flag (AVF) is set, indicating range capacity overflow.
- ****If E = 0:**** The transfer is successful without range errors.